Python Complete Guide

Version 1.0 - PrepAI Knowledge Base

1. Introduction

Python is a high-level, interpreted, object-oriented programming language known for its simplicity and extensive libraries.

2. Features

Easy syntax, interpreted, cross-platform, dynamically typed, open source, large ecosystem.

3. Data Types

int, float, str, bool, list, tuple, set, dict.

4. Control Flow

if, elif, else, for, while, break, continue, pass.

5. Functions

Defining functions, arguments, return values, lambda functions, recursion.

6. Object-Oriented Programming

Classes, objects, inheritance, polymorphism, encapsulation, abstraction.

7. Exception Handling

try, except, else, finally, raise, custom exceptions.

8. File Handling

Reading and writing text files using open(), with statement, CSV basics.

9. Modules & Packages

import, from ... import, pip, virtual environments.

10. Popular Libraries

NumPy, Pandas, Matplotlib, Scikit-learn, Flask, FastAPI.

11. Python Interview Questions

- Why is Python popular?
- Difference between list and tuple?
- What is a dictionary?
- Explain *args and **kwargs.
- What are decorators?
- What is a generator?
- What is list comprehension?
- What is a virtual environment?
- Difference between deep copy and shallow copy?

• Explain GIL.

12. Practice Programs

Palindrome, Fibonacci, Prime numbers, File reader, CRUD application, REST API using FastAPI.

13. Best Practices

Follow PEP 8, use meaningful names, write reusable functions, use virtual environments and exception handling.